

GENERAL CHEMISTRY

Principles and Modern Applications

TENTH EDITION

PETRUCCI

HERRING

MADURA

BISSONNETTE

Gases

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CONTENTS

1. Properties of Gases
2. The Simple Gas Laws
3. Combining the Gas Laws
4. Ideal Gas Equation
5. Gases in Chemical Reactions
6. Mixtures of Gases
7. Kinetic Molecular Theory
8. Gas Properties Relating to the Kinetic Molecular Theory

Properties of Gases



◀ **FIGURE 6-1**
The gaseous states of three halogens (group 17)
The greenish yellow gas is $\text{Cl}_2(\text{g})$; the brownish red gas is $\text{Br}_2(\text{g})$ above a small pool of liquid bromine; the violet gas is $\text{I}_2(\text{g})$ in contact with grayish-black solid iodine. Most other common gases, such as H_2 , O_2 , N_2 , CO , and CO_2 , are colorless.

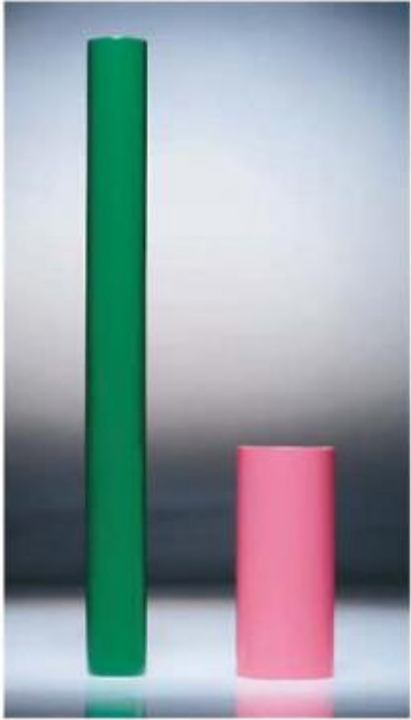
- Gases differ from solids and liquids in several respects: For example, a gas expands spontaneously to fill its container. Consequently, the volume of a gas equals the volume of the container which is held.
- Gases form homogeneous mixtures with each other. Ex: atmosphere. When water and gasoline are mixed, the two liquids remain as separate layers. However, the water vapor and gasoline vapors form homogeneous gas mixture.
- Each molecule behaves as though others are not present. Individual molecules are relatively far apart.
- We cannot see individual particles, although we can see the bulk when it is colored (Fig 6-1).

Four properties determine the physical behavior of a gas: the amount of the gas (in moles) and the volume, temperature, and pressure of the gas. If we know any three of these, we can usually calculate the value of the remaining one by using a mathematical equation called an *equation of state* such as the ideal gas equation.

COMPRESSIBILITY: Gases are easily compressed into a smaller volume. Compressibility is a measure of how the volume of matter decreases under pressure.

PRESSURE

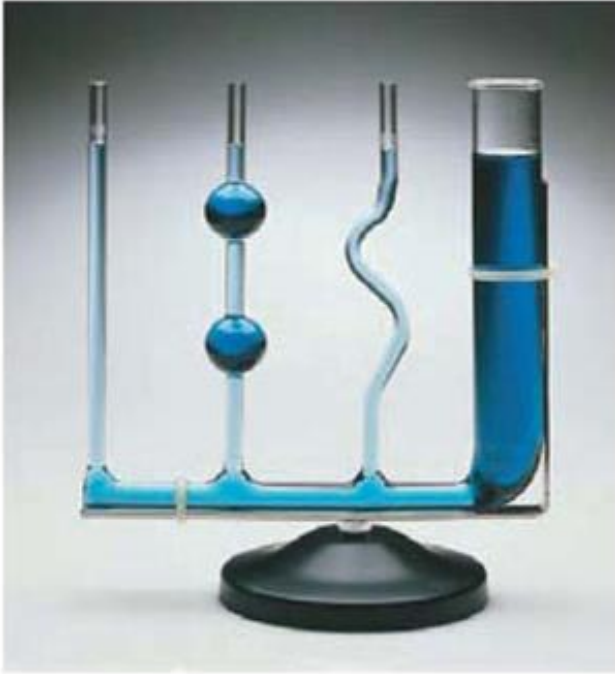
Pressure is defined as a force per unit area, that is, a force divided by the area over which the force is distributed. Figure 6-2 illustrates the idea of pressure exerted by a solid.



▲ FIGURE 6-2
Illustrating the pressure exerted by a solid
The two cylinders have the same mass and exert the same force on the supporting surface ($F = g \times m$). The tall, thin one has a smaller area of contact, however, and exerts a greater pressure ($P = F/A$).

$$P \text{ (Pa)} = \frac{F(N)}{A(m^2)} \quad (1)$$

LIQUID PRESSURE



▲ FIGURE 6-3
The concept of liquid pressure

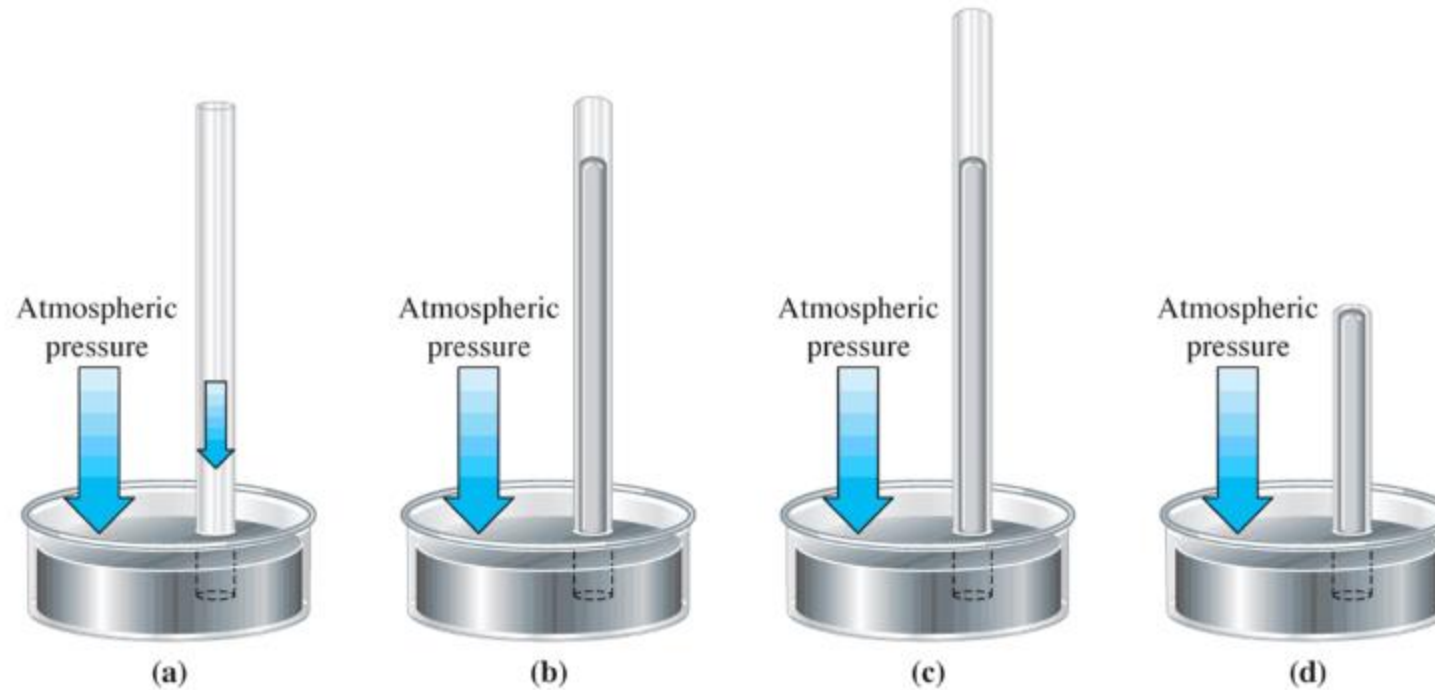
Because it is difficult to measure the total force exerted by gas molecules, it is also difficult to apply equation (1) to gases.

The pressure of a gas is usually measured indirectly, by comparing it with a liquid pressure.

Figure 6-3 illustrates the concept of liquid pressure and suggests that the pressure of a liquid depends only on the height of the liquid column and the density of the liquid.

$$P = \frac{F}{A} = \frac{W}{A} = \frac{gxm}{A} = \frac{gxVxd}{A} = \frac{gxhxAd}{A} = gxhxd$$

BAROMETRIC PRESSURE



▲ FIGURE 6-4
Measurement of atmospheric pressure with a mercury barometer

Evangelista Torricelli constructed the device pictured in Figure 6-4 to measure the pressure exerted by the atmosphere. This device is called a **barometer**.

In the open-end tube (Fig. 6-4a), the atmosphere exerts the same pressure on the surface of the mercury both inside and outside the tube, and the liquid levels are equal.

Inside the closed-end tube (Fig. 6-4b), there is no air above the mercury (only a trace of mercury vapor).

The atmosphere exerts a force on the surface of the mercury in the outside container. This force is transmitted through the liquid, holding up the mercury column within the tube. The column exerts a downward pressure that depends on its height and the density of Hg(l).

The height of mercury in a barometer provides a measure of **barometric pressure**. Barometric pressures may be expressed in a unit called **millimeter of mercury (mmHg)**, defined as the pressure exerted by a column of mercury that is exactly 1 mm in height when the density of mercury is equal to 13.5951 g/cm^3 (0°C) and gravity is equal to 9.80655 m/s^2 .

$$\mathbf{1 \text{ atm} = 760 \text{ Torr} \approx 760 \text{ mmHg}}$$

EXAMPLE 6-1 Comparing Liquid Pressures

What is the height of a column of water that exerts the same pressure as a column of mercury 76.0 cm (760 mm) high?

Analyze

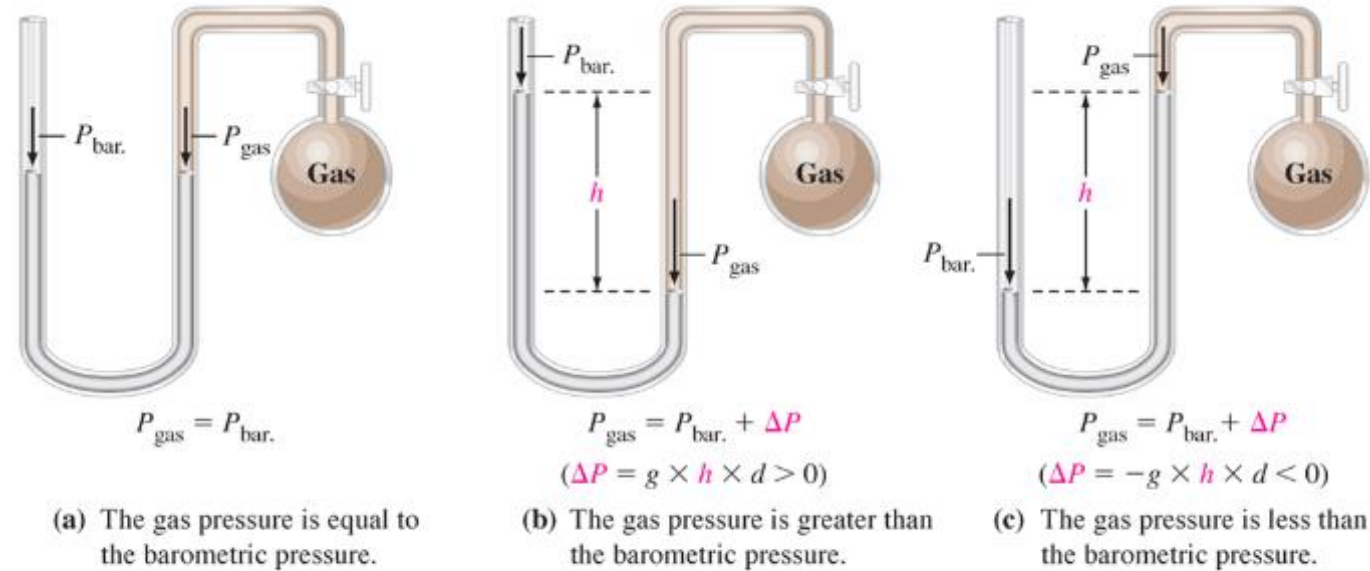
Equation (6.2) shows that, for a given liquid pressure, the column height is inversely proportional to the liquid density. The lower the liquid density, the greater the height of the liquid column. Mercury is 13.6 times as dense as water (13.6 g/cm^3 versus 1.00 g/cm^3). If columns of water and mercury exert the same pressure, then the column of water is 13.6 times as high as the column of mercury.

Solve

Although we have already reasoned out the answer, we can arrive at the same conclusion by applying equation (6.2) twice, and then setting the two pressures equal to each other. Equation (6.2) can be used to describe the pressure of the mercury column of known height and the pressure of the water column of unknown height. Then we can set the two pressures equal to each other.

$$\begin{aligned}\text{pressure of Hg column} &= g \times h_{\text{Hg}} \times d_{\text{Hg}} = g \times 76.0 \text{ cm} \times 13.6 \text{ g/cm}^3 \\ \text{pressure of H}_2\text{O column} &= g \times h_{\text{H}_2\text{O}} \times d_{\text{H}_2\text{O}} = g \times h_{\text{H}_2\text{O}} \times 1.00 \text{ g/cm}^3 \\ g \times h_{\text{H}_2\text{O}} \times 1.00 \text{ g/cm}^3 &= g \times 76.0 \text{ cm} \times 13.6 \text{ g/cm}^3 \\ h_{\text{H}_2\text{O}} &= 76.0 \text{ cm} \times \frac{13.6 \text{ g/cm}^3}{1.00 \text{ g/cm}^3} = 1.03 \times 10^3 \text{ cm} = 10.3 \text{ m}\end{aligned}$$

Manometers



▲ FIGURE 6-5
Measurement of gas pressure with an open-end manometer

Fig 6.5 shows the principle of open-end manometer. When the gas pressure and atmospheric pressure are equal, the heights of the mercury columns in the two arms are equal. A difference in the height of two arms signifies a difference between the gas pressure and barometric pressure.

TABLE 6.1 Some Common Pressure Units

Atmosphere	atm	
Millimeter of mercury	mmHg	$1 \text{ atm} \approx 760 \text{ mmHg}$
Torr	Torr	$= 760 \text{ Torr}$
Pascal	Pa	$= 101,325 \text{ Pa}$
Kilopascal	kPa	$= 101.325 \text{ kPa}$
Bar	bar	$= 1.01325 \text{ bar}$
Millibar	mbar	$= 1013.25 \text{ mbar}$

EXAMPLE 6-2 Using a Manometer to Measure Gas Pressure

When the manometer in Figure 6-5(c) is filled with liquid mercury ($d = 13.6 \text{ g/cm}^3$), the barometric pressure is 748.2 mmHg, and the difference in mercury levels is 8.6 mmHg. What is the gas pressure P_{gas} ?

Analyze

We must first establish which is greater: the barometric pressure or the gas pressure. In Figure 6-5(c), the barometric pressure forces liquid mercury down the tube toward the gas sample. The barometric pressure is greater than the gas pressure. Thus, $\Delta P = P_{\text{gas}} - P_{\text{bar.}} < 0$.

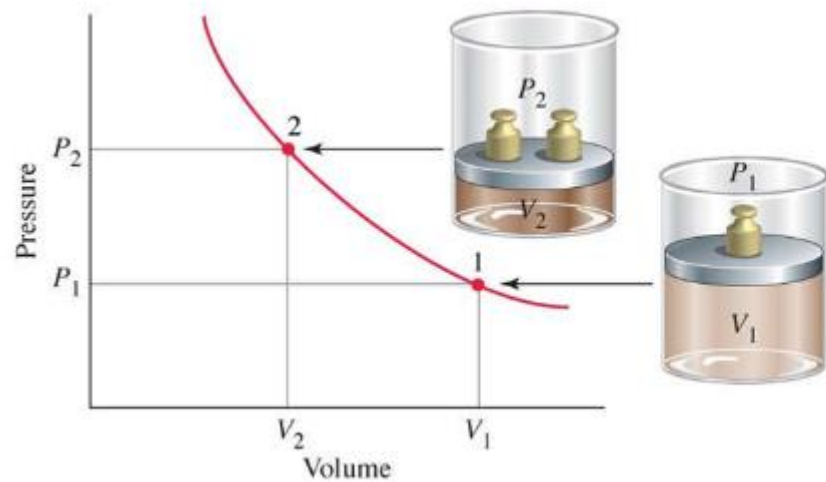
Solve

The gas pressure is less than the barometric pressure. Therefore, we subtract 8.6 mmHg from the barometric pressure to obtain the gas pressure.

$$P_{\text{gas}} = P_{\text{bar.}} + \Delta P = 748.2 \text{ mmHg} - 8.6 \text{ mmHg} = 739.6 \text{ mmHg}$$

The Simple Gas Laws

Boyle's Law: For a fixed amount of gas at a constant temperature, the gas volume is inversely proportional to the gas pressure.



◀ FIGURE 6-6
Relationship between gas volume and
pressure—Boyle's law

$$P \propto 1/V$$
$$P_f V_f = P_i V_i$$

Example: Calculate the volume occupied by a gas when the pressure changes. Consider the 50.0-L gas cylinder of oxygen mentioned in the chapter opening. The pressure of gas in the cylinder is 15.7 atm at 21°C. What volume of oxygen can you get from the cylinder at 21°C if the atmospheric pressure is 1.00 atm?

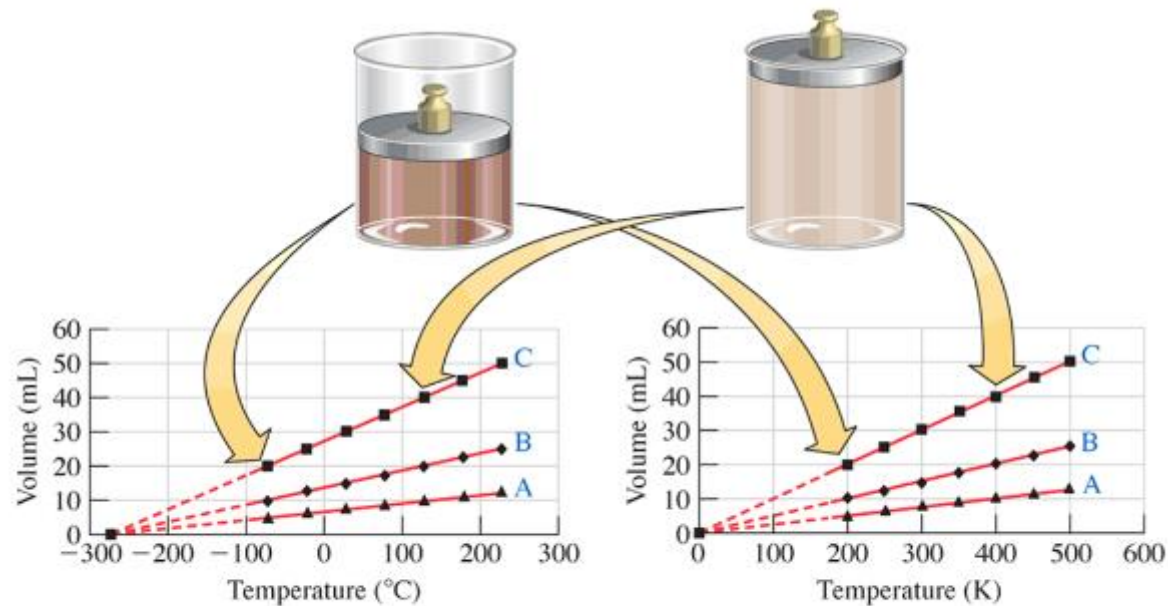
$$P_f V_f = P_i V_i$$

$$V_f 50.0 \text{ L} \times \frac{15.7 \text{ atm}}{1.00 \text{ atm}} = 785 \text{ L}$$

$$785 - 50.0 = 735 \text{ L}$$

The Simple Gas Laws

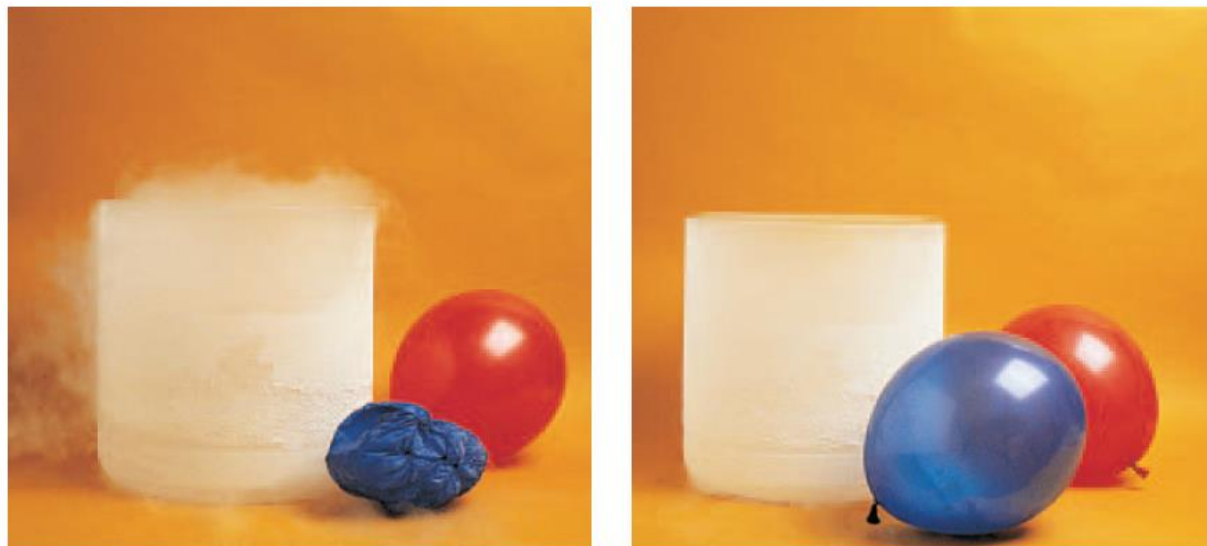
Charles's Law: The pressure is held constant at 1 atm while the temperature is varied. The volume of gas increases as the temperature is raised and decreases as the temperature is lowered. The relationship is linear.



▲ FIGURE 6-7
Gas volume as a function of temperature

$$V \propto T$$

Charles's Law:



Effect of temperature on volume of a gas

A balloon immersed in liquid nitrogen shrinks because the air inside contracts in volume. When the balloon is removed from the liquid nitrogen, the air inside warms up and the balloon expands to its original size.

Charles's Law:

The temperature at which the volume of a hypothetical* gas becomes zero is the absolute zero of temperature: $-273.15\text{ }^{\circ}\text{C}$ on the Celsius scale or 0 K on the **absolute**, or **Kelvin**, scale. The relationship between the Kelvin temperature, T , and the Celsius temperature, t , is shown below in equation:

$$T(\text{K}) = t(^{\circ}\text{C}) + 273.15$$

The volume of a fixed amount of gas at constant pressure is directly proportional to the Kelvin (absolute) temperature.

Charles's Law:

Example: Earlier we found that the total volume of oxygen that can be obtained from a particular tank at 1.00 atm and 21°C is 785 L (including the volume remaining in the tank). What would be this volume of oxygen if the temperature had been 28°C?

$$T_i = 21 + 273 = 294 \text{ K}$$

$$T_f = 28 + 273 = 301 \text{ K}$$

$$V_i = 785 \text{ L}$$

$$V_f = ?$$

$$V_f = V_i \times \frac{T_f}{T_i} = 785 \text{ L} \times \frac{301 \text{ K}}{294 \text{ K}} = 804 \text{ L}$$

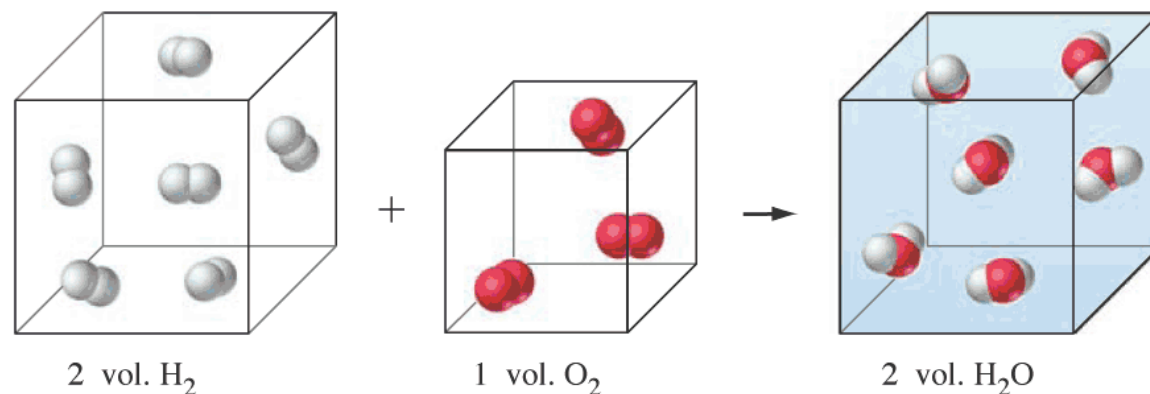
Avogadro's Law:

1. Equal volumes of different gases compared at the same temperature and pressure contain equal numbers of molecules.
2. Equal numbers of molecules of different gases compared at the same temperature and pressure occupy equal volumes.

A relationship that follows from *Avogadro's hypothesis*, often called **Avogadro's law**, is as follows.

If the number of moles of gas (n) is doubled, the volume doubles, and so on.

The molar volume of a 1 mol gas is approximately 22.414 L at 0 °C and 1 atm (STP)



$$V \propto n$$

▲ FIGURE 6-8
Formation of water—actual observation and Avogadro's hypothesis

TABLE 6.2 Densities and Molar Volumes of Various Gases

Gas	Molar Mass, g mol^{-1}	Density (at STP), g L^{-1}	Molar Volume, ^a L mol^{-1}	
			(at STP)	(at 0°C, 1 atm)
H ₂	2.01588	8.87104×10^{-2}	22.724	22.427
He	4.00260	0.17615	22.722	22.425
Ideal gas	—	—	22.711	22.414
N ₂	28.0134	1.23404	22.701	22.404
CO	28.0101	1.23375	22.696	22.399
O ₂	31.9988	1.41034	22.689	22.392
CH ₄	16.0425	0.70808	22.656	22.360
NF ₃	71.0019	3.14234	22.595	22.300
CO ₂	44.0095	1.95096	22.558	22.263
N ₂ O	44.0128	1.95201	22.550	22.255
C ₂ H ₆	30.0690	1.33740	22.483	22.189
NH ₃	17.0352	0.76139	22.374	22.081
SF ₆	146.0554	6.52800	22.374	22.081
C ₃ H ₈	44.0956	1.98318	22.235	21.944
SO ₂	64.064	2.89190	22.153	21.863

Source: The densities are from the National Institute of Standards and Technology (NIST) *Chemistry WebBook*, available online at <http://webbook.nist.gov/chemistry/>.

^aThe molar volume is equal to the molar mass divided by the density. The molar volume at 0 °C and 1 atm is obtained by dividing the molar volume at STP by 1.01325.

Combining the Gas Laws: The Ideal Gas Equation and the General Gas Equation

Boyle's law ($V \propto 1/P$) and Charles's law ($V \propto T$) can be combined and expressed in a single statement: the volume occupied by a given amount of gas is proportional to the absolute temperature divided by the pressure ($V \propto T/P$). You can write this as an equation.

$$V = \text{constant} \times \frac{T}{P} \quad \text{or} \quad \frac{PV}{T} = \text{constant} \quad (\text{for a given amount of gas})$$

Consider a problem in which you wish to calculate the final volume of a gas when the pressure and temperature are changed. PV/T is constant for a given amount of gas, so you can write

$$\frac{P_f V_f}{T_f} = \frac{P_i V_i}{T_i}$$

which rearranges to

$$V_f = V_i \times \frac{P_i}{P_f} \times \frac{T_f}{T_i}$$

The Ideal Gas Equation

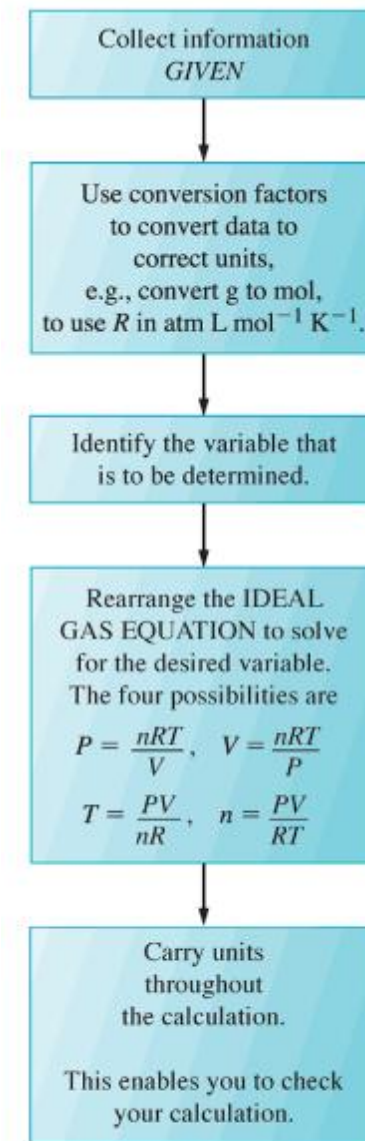
Boyle's Law: $V \propto \frac{1}{P}$ (constant n, T)

Charles's Law: $V \propto T$ (constant n, P)

Avogadro's Law: $V \propto n$ (constant P, T)

$$V = R\left(\frac{nT}{P}\right)$$

$$PV=nRT$$



▲ Applying the ideal gas
equation

EXAMPLE 6-4 Calculating a Gas Volume with the Ideal Gas Equation

What is the volume occupied by 13.7 g $\text{Cl}_2(\text{g})$ at 45 °C and 745 mmHg?

Analyze

This is a relatively straightforward application of the ideal gas equation. We are given an amount of gas (in grams), a pressure (in mmHg), and a temperature (in °C). Before using the ideal gas equation, we must express the amount in moles, the pressure in atmospheres, and the temperature in Kelvin. Include units throughout the calculation to ensure that the final result has acceptable units.

Solve

$$P = 745 \text{ mmHg} \times \frac{1 \text{ atm}}{760 \text{ mmHg}} = 0.980 \text{ atm}$$

$$V = ?$$

$$n = 13.7 \text{ g Cl}_2 \times \frac{1 \text{ mol Cl}_2}{70.91 \text{ g Cl}_2} = 0.193 \text{ mol Cl}_2$$

$$R = 0.08206 \text{ atm L mol}^{-1} \text{ K}^{-1}$$

$$T = 45^\circ\text{C} + 273 = 318 \text{ K}$$

Divide both sides of the ideal gas equation by P to solve for V .

$$\frac{PV}{P} = \frac{nRT}{P} \quad \text{and} \quad V = \frac{nRT}{P}$$

$$V = \frac{nRT}{P} = \frac{0.193 \text{ mol} \times 0.08206 \text{ atm L mol}^{-1} \text{ K}^{-1} \times 318 \text{ K}}{0.980 \text{ atm}} = 5.14 \text{ L}$$

Gas Densities and Molar Mass

To determine the density of a gas, we can start with the density equation, $d=m/V$.

Then we can express the mass of gas as the product of the number of moles of gas and the molar mass: $m=n \times M$

This leads to

$$d = \frac{m}{V} = \frac{n \times M}{V} = \frac{n}{V} \times M$$

n/V equivalent to P/RT

$$d = \frac{m}{V} = \frac{MP}{RT}$$

Example: What is the density of oxygen, O₂, in grams per liter at 25°C and 0.850 atm?

The data given are

Variable	Value
P	0.850 atm
V	1 L (exact value)
T	(25 + 273) K = 298 K
n	?

Therefore,

$$n = \frac{PV}{RT} = \frac{0.850 \text{ atm} \times 1 \text{ L}}{0.0821 \text{ L} \cdot \text{atm}/(\text{K} \cdot \text{mol}) \times 298 \text{ K}} = 0.0347 \text{ mol}$$

Now convert mol O₂ to grams

$$0.0347 \text{ mol O}_2 \times \frac{32.0 \text{ g O}_2}{1 \text{ mol O}_2} = 1.11 \text{ g O}_2$$

Therefore, the density of O₂ at 25°C and 0.850 atm is **1.11 g/L**.

Suppose you wanted the density of Cl₂, instead of O₂, at this T and P . Only the conversion of moles to mass is affected. The previous calculation becomes

$$0.0347 \text{ mol Cl}_2 \times \frac{70.9 \text{ g Cl}_2}{1 \text{ mol Cl}_2} = 2.46 \text{ g Cl}_2$$

The density of Cl₂ at 25°C and 0.850 atm is 2.46 g/L. Note that the density of a gas is directly proportional to its molecular weight.

EXAMPLE 6-7 Determining a Molar Mass with the Ideal Gas Equation

Propylene is an important commercial chemical (about ninth in the amount produced among manufactured chemicals) used in the synthesis of other organic chemicals and in production of plastics (polypropylene). A glass vessel weighs 40.1305 g when clean, dry, and evacuated; it weighs 138.2410 g when filled with water at 25.0 °C (density of water = 0.9970 g/mL) and 40.2959 g when filled with propylene gas at 740.3 mmHg and 24.0 °C. What is the molar mass of propylene?

Analyze

We are given a pressure (in mmHg), a temperature (in °C), and information that will enable us to determine the amount of gas (in grams) and the volume of the vessel. If we express these quantities in Kelvin, atmospheres, moles, and liters, respectively, then we can use equation (6.13), with $R = 0.08206 \text{ atm L K}^{-1} \text{ mol}^{-1}$, to calculate the molar mass of the gas.

Solve

First determine the mass of water required to fill the vessel.

$$\begin{aligned}\text{mass of water to fill vessel} &= 138.2410 \text{ g} - 40.1305 \text{ g} \\ &= 98.1105 \text{ g}\end{aligned}$$

Use the density of water in a conversion factor to obtain the volume of water (and hence, the volume of the glass vessel).

$$\begin{aligned}\text{volume of water (volume of vessel)} &= 98.1105 \text{ g H}_2\text{O} \times \frac{1 \text{ mL H}_2\text{O}}{0.9970 \text{ g H}_2\text{O}} \\ &= 98.41 \text{ mL} = 0.09841 \text{ L}\end{aligned}$$

The mass of the gas is the difference between the weight of the vessel filled with propylene gas and the weight of the empty vessel.

$$\text{mass of gas} = 40.2959 \text{ g} - 40.1305 \text{ g} = 0.1654 \text{ g}$$

The values of temperature and pressure are given.

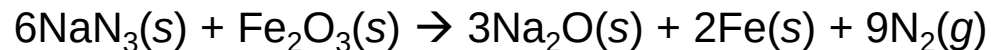
$$\begin{aligned}T &= 24.0 \text{ °C} + 273.15 = 297.2 \text{ K} \\ P &= 740.3 \text{ mmHg} \times \frac{1 \text{ atm}}{760 \text{ mmHg}} = 0.9741 \text{ atm}\end{aligned}$$

Substitute data into the rearranged version of equation (6.13).

$$\begin{aligned}M &= \frac{mRT}{PV} = \frac{0.1654 \text{ g} \times 0.08206 \text{ atm L mol}^{-1} \text{ K}^{-1} \times 297.2 \text{ K}}{0.9741 \text{ atm} \times 0.09841 \text{ L}} \\ &= 42.08 \text{ g mol}^{-1}\end{aligned}$$

Gases in Chemical Reactions

Example: Many such air bags are inflated with nitrogen, N_2 , using the rapid reaction of sodium azide, NaN_3 , and iron(III) oxide, Fe_2O_3 , which is initiated by a spark. The overall reaction is



How many grams of sodium azide would be required to provide 75.0 L of nitrogen gas at 25°C and 748 mmHg?

Here are the available data:

Variable	Value
----------	-------

P	$748 \text{ mmHg} \times \frac{1 \text{ atm}}{760 \text{ mmHg}} = 0.984 \text{ atm}$
-----	--

V	75.0 L
-----	--------

T	$(25 + 273) \text{ K} = 298 \text{ K}$
-----	--

n	?
-----	---

Note that the pressure was converted from mmHg to atmospheres and the temperature from degrees Celsius to kelvins. Rearrange the ideal gas law to obtain n .

$$n = \frac{PV}{RT}$$

Substituting from the data gives you the moles of N_2 .

$$n = \frac{0.984 \text{ atm} \times 75.0 \text{ L}}{0.0821 \text{ L} \cdot \text{atm}/(\text{K} \cdot \text{mol}) \times 298 \text{ K}} = 3.02 \text{ mol}$$

From the moles of N_2 , you use the chemical equation to obtain moles of NaN_3 .

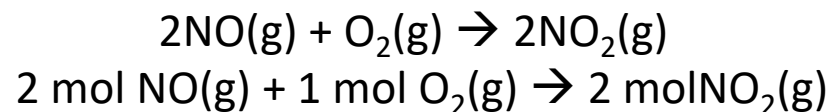
$$3.02 \text{ mol } N_2 \times \frac{6 \text{ mol } NaN_3}{9 \text{ mol } N_2} = 2.01 \text{ mol } NaN_3$$

Now you can calculate the grams of NaN_3 . The molecular weight of NaN_3 is 65.01 amu.

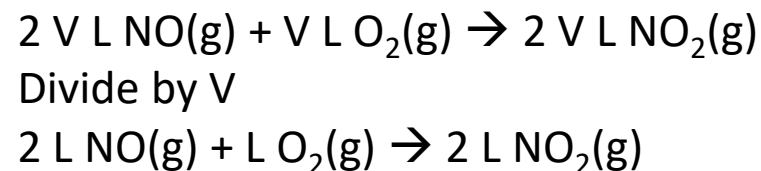
$$2.01 \text{ mol } NaN_3 \times \frac{65.01 \text{ g } NaN_3}{1 \text{ mol } NaN_3} = \mathbf{131 \text{ g } NaN_3}$$

Law of Combining Volumes

If the reactants and products involved in a stoichiometric calculation are gases, sometimes we can use a particularly simple approach. Consider this reaction.

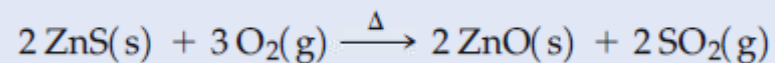


Suppose the gases are compared at the same T and P . Under these conditions, one mole of gas occupies a particular volume, call it V liters; two moles of gas occupy $2V$ liters; and so on.



EXAMPLE 6-10 Applying the Law of Combining Volumes

Zinc blende, ZnS, is the most important zinc ore. Roasting (strong heating) of ZnS in oxygen is the first step in the commercial production of zinc.



What volume of $\text{SO}_2\text{(g)}$ can be obtained from 1.00 L $\text{O}_2\text{(g)}$ and excess ZnS(s)? Both gases are measured at 25 °C and 745 mmHg.

Analyze

The reactant and product being compared are both gases, and both are at the same temperature and pressure. Therefore, we can use the law of combining volumes and treat the coefficients in the balanced chemical equation as if they had units of liters.

Solve

The stoichiometric factor (shown below in blue) converts from L $\text{O}_2\text{(g)}$ to L $\text{SO}_2\text{(g)}$.

$$? \text{ L SO}_2\text{(g)} = 1.00 \text{ L O}_2\text{(g)} \times \frac{2 \text{ L SO}_2\text{(g)}}{3 \text{ L O}_2\text{(g)}} = 0.667 \text{ L SO}_2\text{(g)}$$

Mixtures of Gases

The pressure of each gas is proportional to the number of moles of gas. The total pressure is the sum of the partial pressures of the individual gases.

Dalton's law of partial pressures states that the total pressure of a mixture of gases is the sum of the partial pressures of the components of the mixture. For a mixture of gases, A, B, and so on

$$P_{\text{tot}} = P_A + P_B + \cdots \quad (6.16)$$

In a gaseous mixture of n_A moles of A, n_B moles of B, and so on, the volume each gas would individually occupy at a pressure equal to P_{tot} is

$$V_A = n_A RT / P_{\text{tot}}; V_B = n_B RT / P_{\text{tot}}; \cdots$$

The total volume of the gaseous mixture is

$$V_{\text{tot}} = V_A + V_B \cdots$$

and the commonly used expression *percent by volume* is

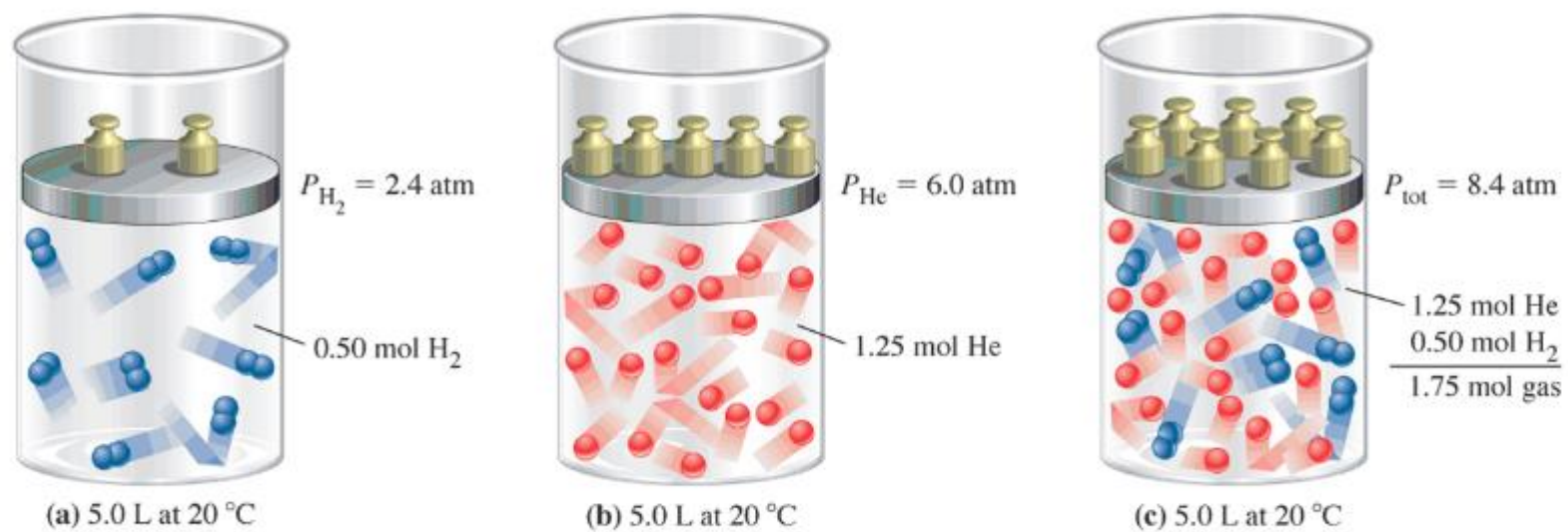
$$\text{volume \% A} = \frac{V_A}{V_{\text{tot}}} \times 100\%; \text{Volume \% B} = \frac{V_B}{V_{\text{tot}}} \times 100\%; \cdots$$

We can derive a particularly useful expression from the following ratios,

$$\frac{P_A}{P_{\text{tot}}} = \frac{n_A (\cancel{RT/V_{\text{tot}}})}{n_{\text{tot}} (\cancel{RT/V_{\text{tot}}})} = \frac{n_A}{n_{\text{tot}}} \quad \text{and} \quad \frac{V_A}{V_{\text{tot}}} = \frac{n_A (\cancel{RT/P_{\text{tot}}})}{n_{\text{tot}} (\cancel{RT/P_{\text{tot}}})} = \frac{n_A}{n_{\text{tot}}}$$

which means that

$$\frac{n_A}{n_{\text{tot}}} = \frac{P_A}{P_{\text{tot}}} = \frac{V_A}{V_{\text{tot}}} = x_A \quad (6.17)$$



▲ FIGURE 6-12
Dalton's law of partial pressures illustrated

Calculating Partial Pressures and Mole Fractions of a Gas in a Mixture

A 1.00-L sample of dry air at 25°C and 786 mmHg contains 0.925 g N₂, plus other gases including oxygen, argon, and carbon dioxide. a. What is the partial pressure (in mmHg) of N₂ in the air sample? b. What is the mole fraction and mole percent of N₂ in the mixture?

a. Each gas in a mixture follows the ideal gas law. To calculate the partial pressure of N₂, you convert 0.925 g N₂ to moles N₂.

$$0.925 \text{ g N}_2 \times \frac{1 \text{ mol N}_2}{28.0 \text{ g N}_2} = 0.0330 \text{ mol N}_2$$

You substitute into the ideal gas law (noting that 25°C is 298 K).

$$\begin{aligned} P_{\text{N}_2} &= \frac{n_{\text{N}_2}RT}{V} \\ &= \frac{0.0330 \text{ mol} \times 0.0821 \text{ L} \cdot \text{atm}/(\text{K} \cdot \text{mol}) \times 298 \text{ K}}{1.00 \text{ L}} \\ &= 0.807 \text{ atm (= 613 mmHg)} \end{aligned}$$

b. The mole fraction of N₂ in air is

$$\text{Mole fraction of N}_2 = \frac{P_{\text{N}_2}}{P} = \frac{613 \text{ mmHg}}{786 \text{ mmHg}} = \mathbf{0.780}$$

Air contains **78.0 mole percent** of N₂.

Collecting Gas Over Liquid

A gas collected through a trough filled with water is said to be *collected over water* and is wet. It is a mixture of two gases: the desired gas and water vapor.

The gas being collected expands to fill the container and exerts its partial pressure, P_{gas} .

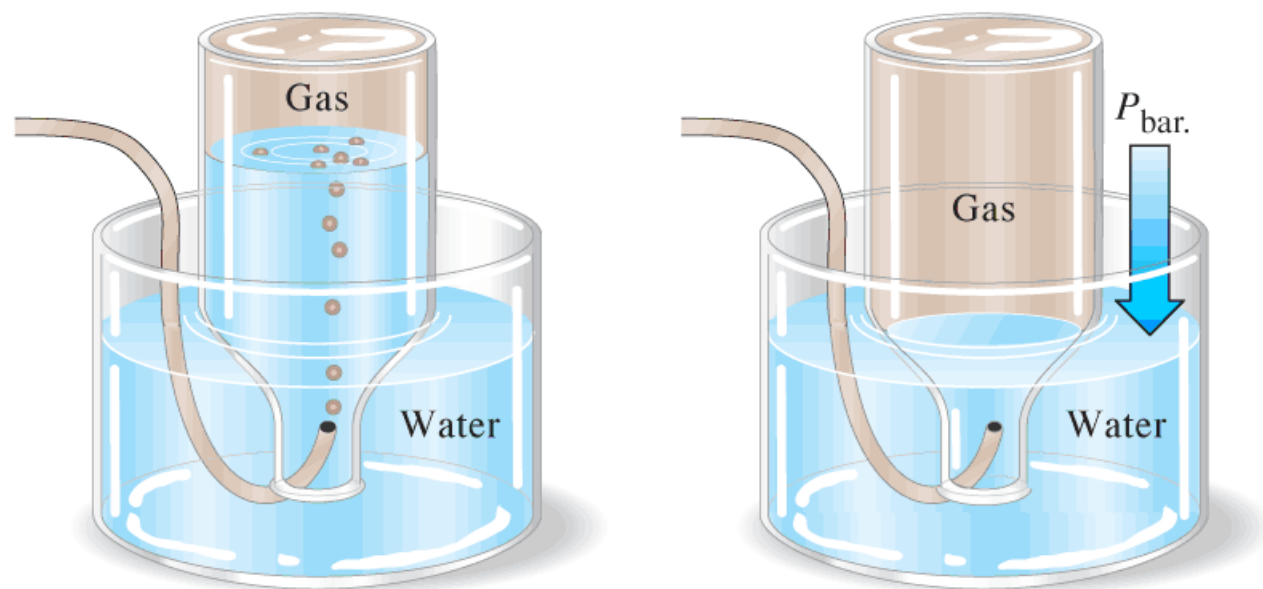
Water vapor, formed by the evaporation of liquid water, also fills the container and exerts a partial pressure, $P_{\text{H}_2\text{O}}$.

The pressure of the water vapor depends only on the temperature of the water, as shown in Table 6.4.

$$P_{\text{tot}} = P_{\text{bar.}} = P_{\text{gas}} + P_{\text{H}_2\text{O}} \quad \text{or} \quad P_{\text{gas}} = P_{\text{bar.}} - P_{\text{H}_2\text{O}}$$

TABLE 6.4 Vapor Pressure of Water at Various Temperatures

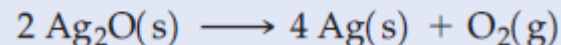
Temperature, °C	Vapor Pressure, mmHg
15.0	12.79
17.0	14.53
19.0	16.48
21.0	18.65
23.0	21.07
25.0	23.76
30.0	31.82
50.0	92.51



▲ FIGURE 6-13
Collecting a gas over water

EXAMPLE 6-13 Collecting a Gas over a Liquid (Water)

In the following reaction, 81.2 mL of $\text{O}_2(\text{g})$ is collected over water at 23°C and barometric pressure 751 mmHg. What mass of $\text{Ag}_2\text{O}(\text{s})$ decomposed? (The vapor pressure of water at 23°C is 21.1 mmHg.)



Analyze

The key concept is that the gas collected is wet, that is, a *mixture* of $\text{O}_2(\text{g})$ and water vapor. Use $P_{\text{bar.}} = P_{\text{O}_2} + P_{\text{H}_2\text{O}}$ to calculate P_{O_2} , and then use the ideal gas equation to calculate the number of moles of O_2 . The following conversions are used to complete the calculation: $\text{mol O}_2 \longrightarrow \text{mol Ag}_2\text{O} \longrightarrow \text{g Ag}_2\text{O}$.

Solve

$$P_{\text{O}_2} = P_{\text{bar.}} - P_{\text{H}_2\text{O}} = 751 \text{ mmHg} - 21.1 \text{ mmHg} = 730 \text{ mmHg}$$

$$P_{\text{O}_2} = 730 \text{ mmHg} \times \frac{1 \text{ atm}}{760 \text{ mmHg}} = 0.961 \text{ atm}$$

$$V = 81.2 \text{ mL} = 0.0812 \text{ L}$$

$$n = ?$$

$$R = 0.08206 \text{ atm L mol}^{-1} \text{ K}^{-1}$$

$$T = 23^\circ\text{C} + 273 = 296 \text{ K}$$

$$n = \frac{PV}{RT} = \frac{0.961 \text{ atm} \times 0.0812 \text{ L}}{0.08206 \text{ atm L mol}^{-1} \text{ K}^{-1} \times 296 \text{ K}} = 0.00321 \text{ mol}$$

From the chemical equation we obtain a factor to convert from moles of O_2 to moles of Ag_2O . The molar mass of Ag_2O provides the final factor.

$$? \text{ g Ag}_2\text{O} = 0.00321 \text{ mol O}_2 \times \frac{2 \text{ mol Ag}_2\text{O}}{1 \text{ mol O}_2} \times \frac{231.7 \text{ g Ag}_2\text{O}}{1 \text{ mol Ag}_2\text{O}} = 1.49 \text{ g Ag}_2\text{O}$$

Kinetic-Molecular Theory of Gases

It is based on the *model* illustrated in Figure 6-14 and outlined as follows.

- A gas is composed of a very large number of extremely small particles in constant, random, straight-line motion.
- Molecules of a gas are separated by great distances. The gas is mostly empty space.
- Molecules collide only fleetingly with one another and with the walls of their container.
- There are assumed to be no forces between molecules
- Individual molecules may gain or lose energy as a result of collisions however, *the total energy remains constant*.



▲ FIGURE 6-14
Visualizing molecular motion

Gas Properties Relating to the Kinetic-Molecular Theory

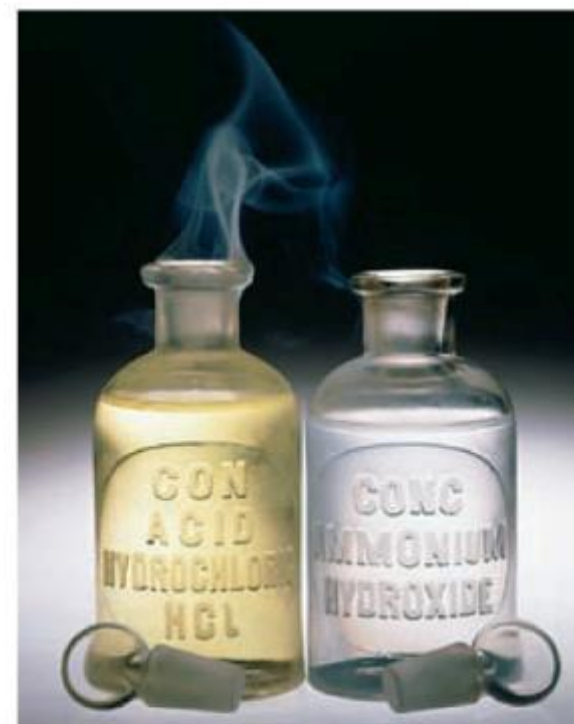
Diffusion is the migration of molecules as a result of random molecular motion.

Effusion, is the escape of gas molecules from their container through a tiny orifice or pinhole.

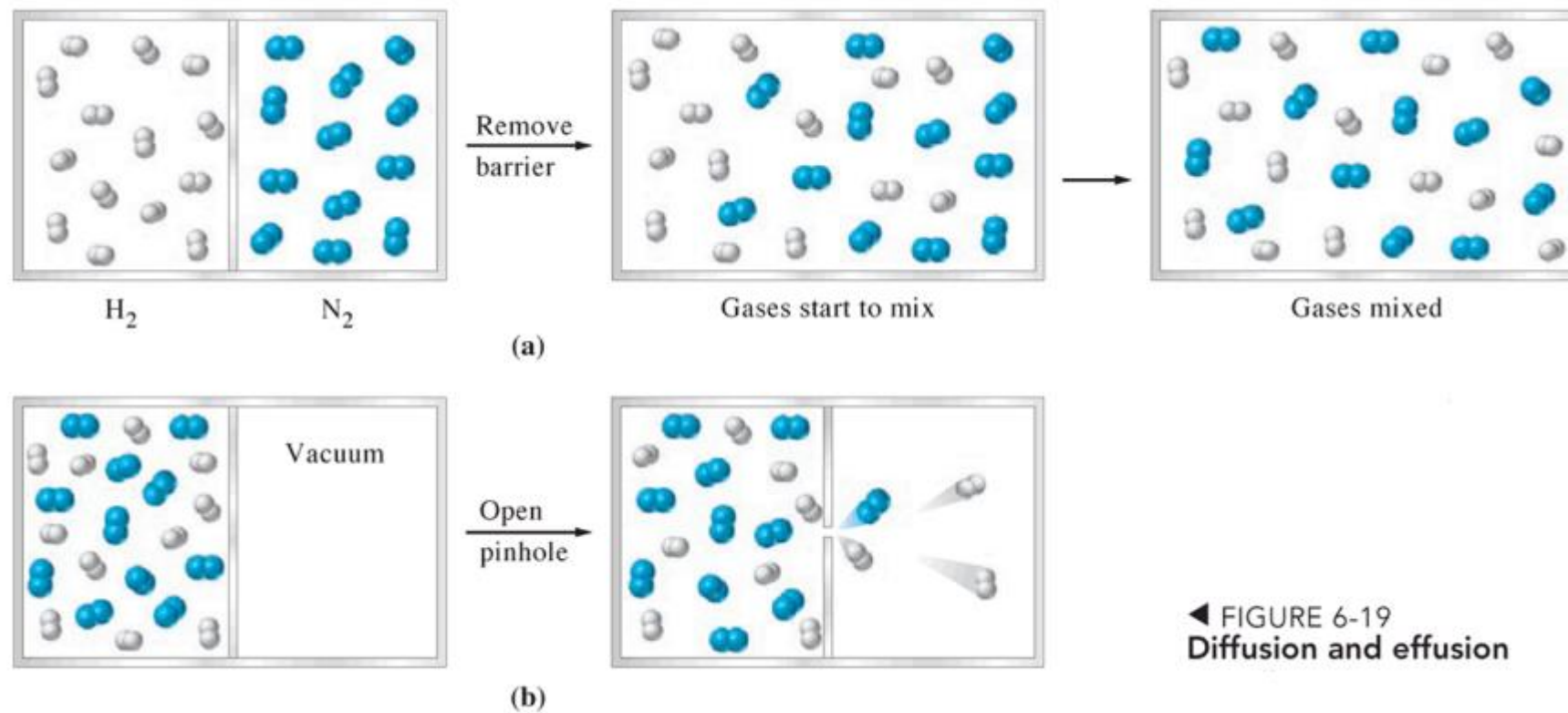
Graham's law of effusion:

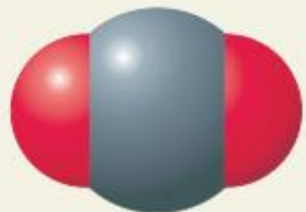
Rate of effusion of molecules $\propto \frac{1}{\sqrt{M_m}}$ (for the same container at constant T and P)

$$\frac{\text{rate of effusion of A}}{\text{rate of effusion of B}} = \frac{(u_{\text{rms}})_A}{(u_{\text{rms}})_B} = \sqrt{\frac{3RT/M_A}{3RT/M_B}} = \sqrt{\frac{M_B}{M_A}}$$



▲ FIGURE 6-18
Diffusion of $\text{NH}_3(\text{g})$
and $\text{HCl}(\text{g})$





CO_2 ; $M_m = 44.0 \text{ g/mol}$



SO_2 ; $M_m = 64.1 \text{ g/mol}$

Calculating the Ratio of Effusion Rates of Gases

Calculate the ratio of effusion rates of molecules of carbon dioxide, CO_2 , and sulfur dioxide, SO_2 , from the same container and at the same temperature and pressure.

The two rates of effusion are inversely proportional to the square roots of their molar masses, so you can write

$$\frac{\text{Rate of effusion of } \text{CO}_2}{\text{Rate of effusion of } \text{SO}_2} = \sqrt{\frac{M_m(\text{SO}_2)}{M_m(\text{CO}_2)}}$$

where $M_m(\text{SO}_2)$ is the molar mass of SO_2 (64.1 g/mol) and $M_m(\text{CO}_2)$ is the molar mass of CO_2 (44.0 g/mol). Substituting these molar masses into the formula gives

$$\frac{\text{Rate of effusion of } \text{CO}_2}{\text{Rate of effusion of } \text{SO}_2} = \sqrt{\frac{64.1 \text{ g/mol}}{44.0 \text{ g/mol}}} = \mathbf{1.21}$$

In other words, carbon dioxide effuses 1.21 times faster than sulfur dioxide (because CO_2 molecules move 1.21 times faster on average than SO_2 molecules).

EXAMPLE 6-16 Relating Effusion Times and Molar Masses

A sample of Kr(g) escapes through a tiny hole in 87.3 s. The same amount of an unknown gas escapes in 42.9 s under identical conditions. What is the molar mass of the unknown gas?

Analyze

Because the unknown gas effuses faster, it must have a smaller molar mass than Kr. Before we set the ratio

$$\frac{\text{effusion time for unknown}}{\text{effusion time for Kr}}$$

equal to $\sqrt{\text{ratio of two molar masses}}$, we must make sure the ratio of molar masses is smaller than one. Thus, the ratio of molar masses must be written with the molar mass of the lighter gas (the unknown gas) in the numerator.

Solve

$$\frac{\text{effusion time for unknown}}{\text{effusion time for Kr}} = \frac{42.9 \text{ s}}{87.3 \text{ s}} = \sqrt{\frac{M_{\text{unk}}}{M_{\text{Kr}}}} = 0.491$$

$$M_{\text{unk}} = (0.491)^2 \times M_{\text{Kr}} = (0.491)^2 \times 83.80 = 20.2 \text{ g/mol}$$

Real Gases

A useful measure of how much a gas deviates from ideal gas behavior is found in its compressibility factor. **The compressibility factor** of a gas is the Ratio **PV/nRT**.

From the ideal gas equation we see that for an ideal gas, **PV/nRT=1**.

Values of the compressibility factor are given in Table 6.5 for a variety of gases at 300 K and 10 bar.

$$\left(P + \frac{n^2a}{V^2}\right)(V - nb) = nRT$$

TABLE 6.5 van der Waals Constants and Compressibility Factors (at 10 bar and 300 K) for Various Gases

Gas	van der Waals Constants		Compressibility Factor
	$a, \text{bar L}^2 \text{mol}^{-2}$	$b, \text{L mol}^{-1}$	
H ₂	0.2452	0.0265	1.006
He	0.0346	0.0238	1.005
Ideal gas	0	0	1
N ₂	1.370	0.0387	0.998
CO	1.472	0.0395	0.997
O ₂	1.382	0.0319	0.994
CH ₄	2.303	0.0431	0.983
NF ₃	3.58	0.0545	0.965
CO ₂	3.658	0.0429	0.950
N ₂ O	3.852	0.0444	0.945
C ₂ H ₆	5.580	0.0651	0.922
NH ₃	4.225	0.0371	0.887
SF ₆	5.580	0.0651	0.880
C ₃ H ₈	9.39	0.0905	a
SO ₂	7.857	0.0879	a

Source: van der Waals constants are from the *CRC Handbook of Chemistry and Physics*, 83rd ed., David R. Lide (ed.), Boca Raton, FL: Taylor & Francis Group, 2002. Compressibility factors are calculated by using data from the National Institute of Standards and Technology (NIST) Chemistry WebBook, available online at <http://webbook.nist.gov/chemistry/>.

^aAt 10 bar and 300 K, C₃H₈ and SO₂ are liquids.

EXAMPLE 6-17 Using the van der Waals Equation to Calculate the Pressure of a Gas

Use the van der Waals equation to calculate the pressure exerted by 1.00 mol $\text{Cl}_2(\text{g})$ confined to a volume of 2.00 L at 273 K. The value of $a = 6.49 \text{ L}^2 \text{ atm mol}^{-2}$, and that of $b = 0.0562 \text{ L mol}^{-1}$.

Analyze

This is a straightforward application of equation (6.26). It is important to include units to make sure the units cancel out properly.

Solve

Solve equation (6.26) for P .

$$P = \frac{nRT}{V - nb} - \frac{n^2a}{V^2}$$

Then substitute the following values into the equation.

$$n = 1.00 \text{ mol}; V = 2.00 \text{ L}; T = 273 \text{ K}; R = 0.08206 \text{ atm L mol}^{-1} \text{ K}^{-1}$$

$$n^2a = (1.00)^2 \text{ mol}^2 \times 6.49 \frac{\text{L}^2 \text{ atm}}{\text{mol}^2} = 6.49 \text{ L}^2 \text{ atm}$$

$$nb = 1.00 \text{ mol} \times 0.0562 \text{ L mol}^{-1} = 0.0562 \text{ L}$$

$$P = \frac{1.00 \text{ mol} \times 0.08206 \text{ atm L mol}^{-1} \text{ K}^{-1} \times 273 \text{ K}}{(2.00 - 0.0562) \text{ L}} - \frac{6.49 \text{ L}^2 \text{ atm}}{(2.00)^2 \text{ L}^2}$$

$$P = 11.5 \text{ atm} - 1.62 \text{ atm} = 9.9 \text{ atm}$$